

CPEN 416 / EECE 5710

Lecture 13: QFT and quantum phase estimation

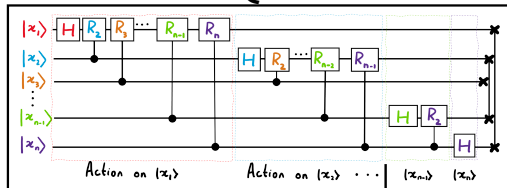
Thursday 23 October 2025

Announcements

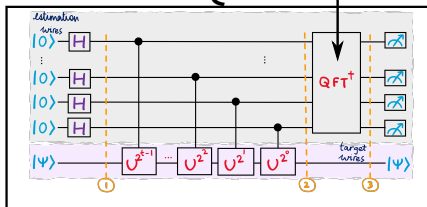
- Quiz 6 on Tuesday
- Tutorial Assignment 2 due tomorrow 23:59
- Tutorial Assignment 3 next week
- Practice Assignment 3 distributed tomorrow

Reminder: where are we going?

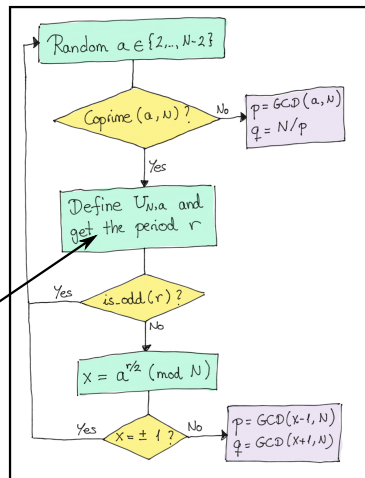
1. QFT



2. QPE



3. Shor



Last time

We introduced the quantum Fourier transform, which is analogous to the inverse discrete Fourier transform.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

$$QFT = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{pmatrix}$$

where for n qubits, $N = 2^n$, and $\omega = e^{2\pi i/N}$

Last time

We worked out the result of the QFT for a few small cases:

0. $x_1, x_2, x_3, \dots$

$$= \frac{x_1}{2} + \frac{x_2}{2^2} + \frac{x_3}{2^3} + \dots$$

$$QFT|x_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_1}|1\rangle)$$

fractional binary



$$QFT|x_1x_2\rangle = \frac{1}{2} \left((|0\rangle + e^{2\pi i 0.x_2}|1\rangle) \otimes (|0\rangle + e^{2\pi i 0.x_1x_2}|1\rangle) \right)$$

$$0.x_2 = \frac{x_2}{2}$$

Example: verify with $QFT|01\rangle$

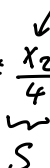
$$\begin{aligned} QFT|01\rangle &= \frac{1}{2} \left(|0\rangle + e^{2\pi i \frac{0}{2}} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i (\frac{0}{2} + \frac{1}{4})} |1\rangle \right) \\ &= \frac{1}{2} \left(|0\rangle - 1 \cdot |1\rangle \right) \otimes \left(|0\rangle + i |1\rangle \right) \\ &= \frac{1}{2} \left(|0\rangle - |1\rangle \right) \left(|0\rangle + i |1\rangle \right) \end{aligned}$$



Last time

$$QFT|x_1x_2x_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_3}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_2x_3}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_1x_2x_3}|1\rangle)$$

$$\frac{x_1}{2} \tau \frac{x_2}{4} \tau \frac{x_3}{8}$$





General expression:

$$|x\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

$$= \frac{(|0\rangle + e^{2\pi i 0.x_n}|1\rangle) (|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle) \dots (|0\rangle + e^{2\pi i 0.x_1 \dots x_n}|1\rangle)}{\sqrt{N}}$$

Learning outcomes

- Construct a circuit for the quantum Fourier transform and analyze its resource usage
- Outline the algorithmic steps of the QPE subroutine
- Implement QPE in PennyLane and estimate the phase of small unitary matrix

A circuit for the QFT

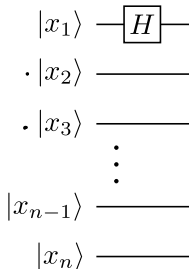
$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \cdots (|0\rangle + e^{2\pi i 0 \cdot x_1 \cdots x_n} |1\rangle)}{\sqrt{N}}$$

Starting with the state

$$|x\rangle = |x_1 \cdots x_n\rangle,$$

apply a Hadamard to qubit 1:

$$H |x_1 \cdots x_n\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot x_1} |1\rangle) |x_2 \cdots x_n\rangle$$



A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

Apply controlled R_2 from qubit

$2 \rightarrow 1$

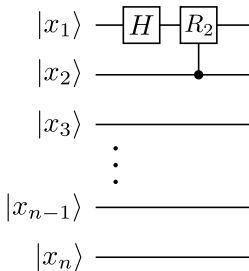
$$\frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot x_1} |1\rangle) |x_2\rangle$$

Controlled R_2

$$\frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot x_1 x_2} |1\rangle) |x_2\rangle$$

$$R_2 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \Rightarrow$$

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^k}} \end{pmatrix}$$



A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

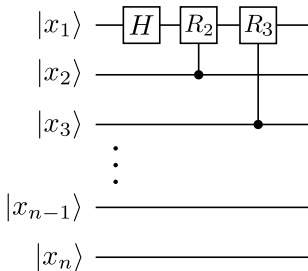
Apply controlled R_3 from qubit

$3 \rightarrow 1$

$$\frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot x_1 x_2} |1\rangle) |x_2\rangle |x_3\rangle$$

$\downarrow R_3 = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/8} \end{pmatrix}$

$$\frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot x_1 x_2 x_3} |1\rangle) |x_2\rangle |x_3\rangle$$

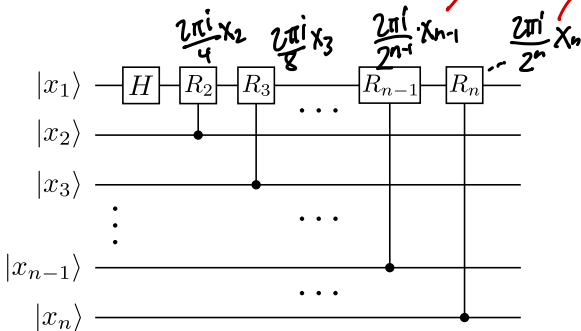


$|x_1 x_2 \dots x_n\rangle$

A circuit for the QFT

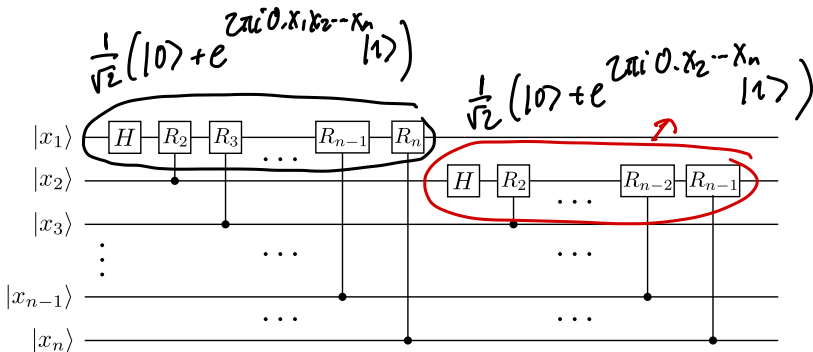
$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

Apply a controlled R_4 from 4 $\rightarrow$ 1, etc. up to the n -th qubit to get



A circuit for the QFT

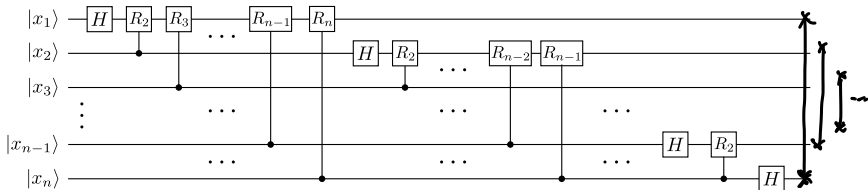
Similar thing with second qubit: apply H then controlled rotations from qubits 3 to n to get



A circuit for the QFT

Do this for all qubits to get that big ugly state from earlier:

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

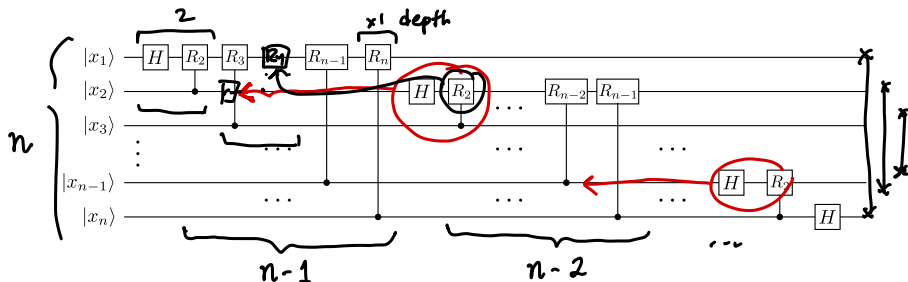


Qubit order is backwards - easily fixed with SWAP gates.

Quantum Fourier transform

Exercise: What are the gate counts and depth of this circuit?

- n Hadamards
 - $\lfloor \frac{n}{2} \rfloor$ SWAPs
 - $\frac{n(n-1)}{2}$ controlled rots.
 - $2n$ depth
- } $\frac{n^2}{2}$ 2qubit gates



Eigenvalues of unitary matrices

Eigenvalues of unitary matrices are complex numbers with magnitude 1, i.e.,

$$U |k\rangle = \lambda_k |k\rangle \rightarrow \lambda_k = e^{2\pi i \theta_k}$$

eigenstate

where θ_k is a phase, $|\theta_k| \leq 1$. (See proof at end of slides)

What if we want to *estimate* an unknown θ_k ?

$$U |k\rangle = e^{2\pi i \theta_k} |k\rangle$$

↑
unobservable global phase! ☹

Quantum phase estimation

Given unitary U and eigenvector $|k\rangle$, estimate θ_k such that

$$U|k\rangle = e^{2\pi i\theta_k}|k\rangle$$

Must determine:

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

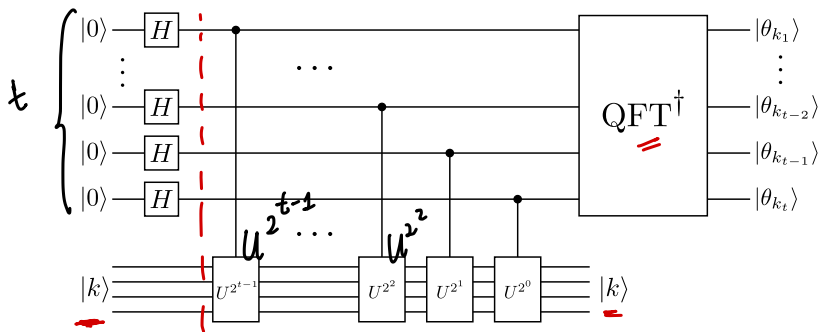
(You can explore the last two in practice assignment 3!)

Quantum phase estimation

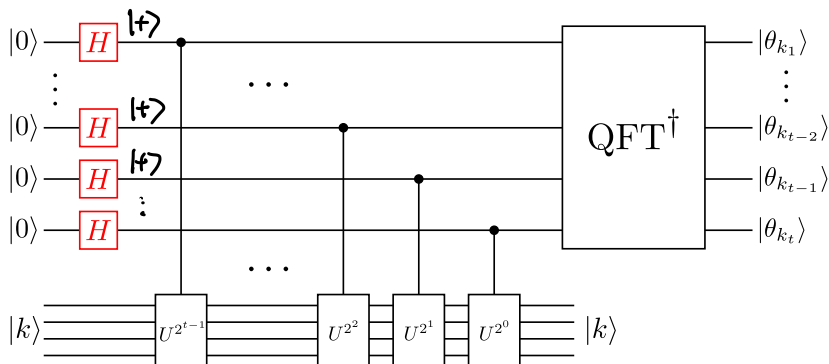
Let U be an n -qubit unitary; $|k\rangle$ is an n -qubit eigenstate.

Assume θ_k can be represented *exactly* using t bits:

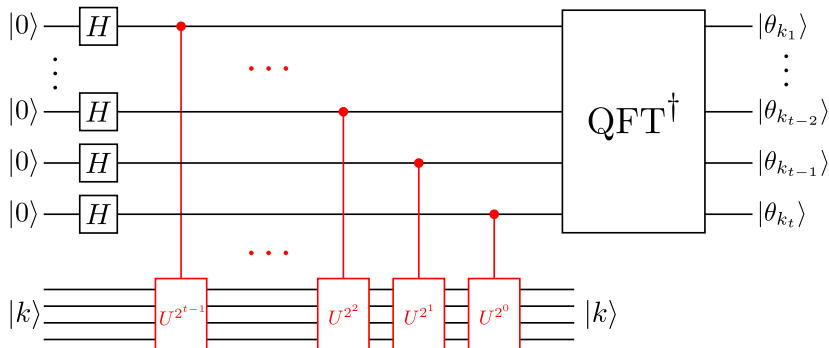
$$\theta_k = 0.\theta_{k_1} \cdots \theta_{k_t}$$



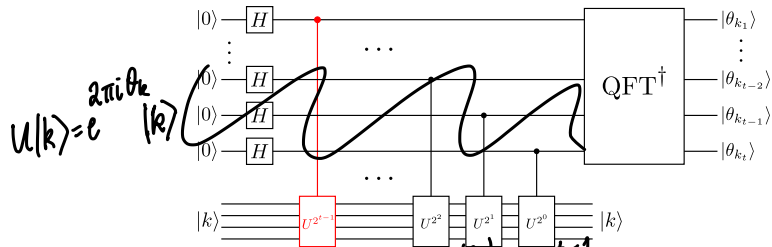
Quantum phase estimation: step 1



Quantum phase estimation: step 2

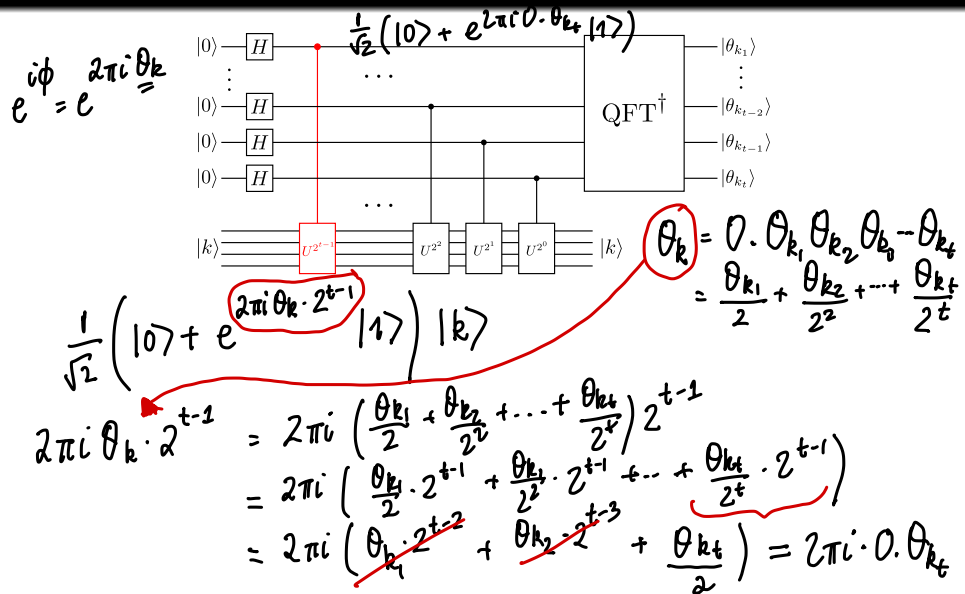


Quantum phase estimation: step 2

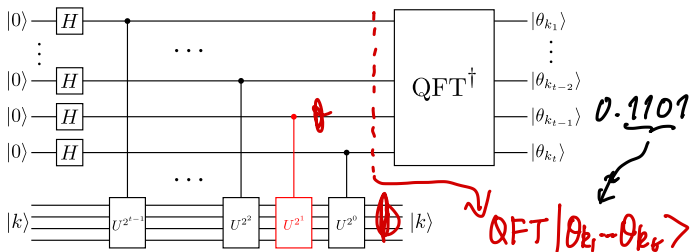


$$\begin{aligned}
 \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \underbrace{|k\rangle}_{\text{target}} &\xrightarrow{C = U^{2^{t-1}}} \frac{1}{\sqrt{2}} |0\rangle |k\rangle + \frac{1}{\sqrt{2}} |1\rangle (e^{2\pi i \theta_k 2^{t-1}} |k\rangle) \\
 &= \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \theta_k \cdot 2^{t-1}} |1\rangle \right) |k\rangle
 \end{aligned}$$

Quantum phase estimation: step 2



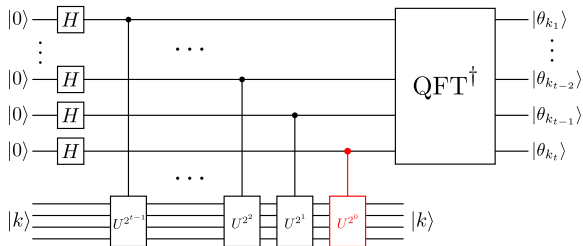
Quantum phase estimation: step 2



Check second-last qubit (ignore the others)

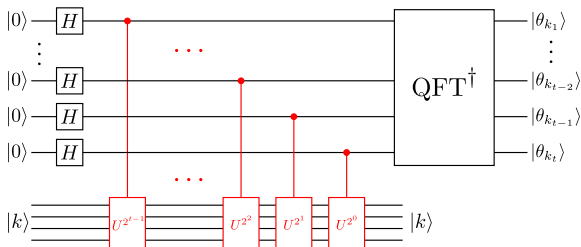
$$\begin{aligned}
 \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right) |k\rangle &\longrightarrow \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \theta_k \cdot 2} |1\rangle \right) |k\rangle \\
 &= \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 0. \theta_{k_2} \theta_{k_3} \dots \theta_{k_t}} |1\rangle \right) |k\rangle
 \end{aligned}$$

Quantum phase estimation: step 2



Can show in the same way for the last qubit (ignore others)

Quantum phase estimation: step 2



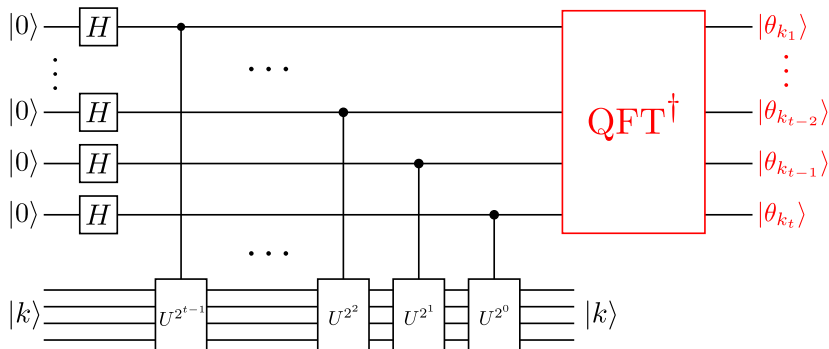
After step 2, we have the state

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_t}}|1\rangle) \cdots \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_2} \cdots \theta_{k_t}}|1\rangle) \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_1} \cdots \theta_{k_t}}|1\rangle)|k\rangle$$

Should look familiar!

Quantum phase estimation: step 3

Measure to learn the bits of θ_k .



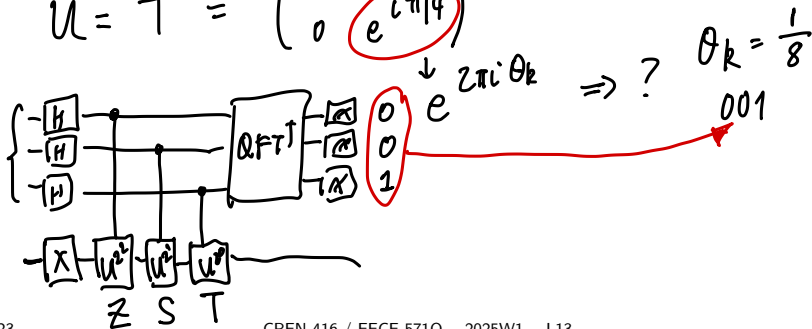
Example: QPE for the T gate

Let's apply QPE to estimate the phase of an eigenstate of T : $|1\rangle$.

- 1 What answer do we expect?
- 2 How many estimation bits?
- 3 What does the circuit look like?

0.25625...

$$U = T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$



Quantum phase estimation

Recall our initial questions... any insights about the second two?

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

Next time

Content:

- Tutorial assignment 3 (QAE with QPE)
- Quiz 6 on Tuesday
- Order finding

Action items:

- ① Work on project (do peer assessment)
- ② Finish tutorial assignment 2

Recommended reading:

- From today: Codebook QFT, QPE, Nielsen & Chuang 5.1-2
- For next class: Codebook SH, Nielsen & Chuang 5.3.1

Eigenvalues of unitary matrices

Fun fact: eigenvalues of unitaries are complex numbers with magnitude 1.

Proof: Let λ_k be the eigenvalue associated with eigenvector $|k\rangle$ of a unitary U :

$$U|k\rangle = \lambda_k|k\rangle$$

Take the conjugate transpose:

$$\langle k| U^\dagger = \langle k| \lambda_k^*$$

Multiply the two sides together:

$$\langle k| U^\dagger U|k\rangle = \langle k| \lambda_k^* \lambda_k |k\rangle$$

$$\langle k|k\rangle = |\lambda_k|^2 \langle k|k\rangle$$

$$1 = |\lambda_k|^2$$